

Oscar Stomberg

Robotics Engineering Student at WPI

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Education

May 2027 **Worcester Polytechnic Institute (WPI), Worcester, MA**
B.S. in Robotics Engineering (anticipated)
Minor: Manufacturing Engineering
Relevant Coursework: Embedded Systems for Robotics, Unified Robotics I-III, Control Engineering, Statics, Dynamics, Materials Science, Object-Oriented Design, Systems Programming, Computer Networks, Full-Stack Software Engineering, Ordinary Differential Equations, Linear Algebra, Probability Theory

Skills

Fabrication

- **3D printing, Laser cutting, CAM** (strong experience with functional part design)
- **Woodworking** (general shop skills)

Hardware & Electronics

- **Electronics prototyping, Soldering, KiCad** (schematic design)
- **Microcontrollers:** Arduino, RP2040, ESP32
- **Sensors:** IMU, encoders, IR rangefinders
- **Protocols:** I2C, SPI, UART

Robotics

- **Forward & inverse kinematics, Sensor fusion, Trajectory planning**
- **Navigation, Proprioception, Computer vision** (OpenCV)

Programming Languages

- **Python:** NumPy, Pandas, Matplotlib, Flask, Requests, OpenCV, etc.
- **C/C++**, particularly for embedded and other low-level applications
- **TypeScript, Node.js, Express, React**
- **PostgreSQL, Prisma, Tailwind CSS + shadcn/ui**
- **MATLAB, Java, JavaScript, Bash**

Infrastructure & DevOps

- **Linux, Docker, Docker Compose, Ansible, Git, GitHub, Jira**
- **Networking:** DNS (Technitium), reverse proxy (Caddy), TLS, SSH, Tailscale
- **Monitoring & backups:** Uptime Kuma, Beszel, Healthchecks, Restic, Rclone, Ntfy

Software

- **CAD:** Proficient in SOLIDWORKS, experienced in Inventor, Fusion360, and FreeCAD
- **Microsoft 365, $\LaTeX$, Zotero, Simulink**
- **JetBrains IDEs, VS Code, Vim/Neovim**

Languages

- **Spanish:** Proficient in reading, writing, listening, and speaking (CEFR B1)

Work Experience

Fall 2025 **Undergraduate Instructional Assistant, WPI Department of Mechanical Engineering**
Assisted in grading and correcting homework assignments and exams for ES 3011: Control Engineering I. Subjects included transfer functions, stability analysis, root-locus plots, classical and modern control engineering.

Projects

Curricular

- Spring 2026 **Software Engineering**, *collaboration with Hanover Insurance*, WPI
Worked as Assistant Lead Developer in a ten person team to apply Agile development methodologies in the creation of a web-based internal content management system for Hanover Insurance. The final application included AI powered full-text search, advanced filtering, tagging, and content organization, a built in content review workflow, user and role management, data analytics, and a customizable dashboard. The application was deployed using AWS, Supabase, and Cloudflare.
- 2025–2026 **Interactive Qualifying Project**, *Waste Management & Recycling*, Akyem Dwenase, Ghana
Developed and introduced a sustainable recycling program for a rural Ghanaian village. Co-designed with stakeholders to create solutions suited to the local culture and context; Conducted field observations and interviews with community members. Increased productivity and decreased occupational hazards for local recycling collectors.
- Spring 2025 **Stem-inism**, *Final Project*
Researched the potential utility and pitfalls of AI chatbots in addressing the national mental health care shortage, presented a set of policy recommendations for ethical development and use.
- Fall 2024 **Introduction to Robotics**, *Team Lab Project*, WPI
Designed, assembled, and programmed a robot with a team to navigate an "orchard", grab, and sort "fruits" into color-coded baskets. Implemented line-following, pathfinding, and computer vision for navigation and sorting.
- Fall 2023 **Great Problems Seminar**, *Climate Change*, WPI
Worked with a team to research adaptation methods for the increasing frequency and intensity of storms in coastal India. Produced a website and presented a poster with our findings.
- Spring 2023 **Engineering Design & Development**, *Capstone Project*, JRHS
Collaborated with peers to develop a smart trash can, aiming to help reduce household waste production. Leveraged our skills in 3D printing, electronics, and programming.

Personal

- 2025–Present **Selfhosting**
Built and maintained a multi-node home server cluster managed with Ansible. Deployed 15+ containerized applications using Docker Compose; Implemented reverse proxy and private DNS; Automated backups with Restic; Integrated robust monitoring stack and home automation systems.
- 2025–2026 **Daily Greeting Generator**
A Python script combining web APIs and other random inputs to generate whimsical daily wake-up announcements using locally hosted LLMs. Served on a Flask web server, containerized with Docker and deployed on local infrastructure; Integrated with Home Assistant automations and Piper TTS.
- Fall 2025 **WPI Class Schedule Importer**
TUI application written in Python to facilitate importing class schedules from WPI's Workday to a calendar format. Parses an input spreadsheet, allows the user to select subsets of terms/classes/sections, produces an iCal file. Provides convenient file dialogs.
- Spring 2025 **Plant Datalogger**
Plant monitoring system developed on the Raspberry Pi Pico W board (RP2040). Written with the C SDK, implements custom logging, NTP, calibration, and error management modules. Plans to integrate MQTT → InfluxDB → Grafana for data storage and visualization, deploy on custom PCB.
- 2021, 2025 **Solar System Simulator**
A standalone Java application to simulate 3D gravitational physics. Implements multiple integration methods and optimized for multi-threaded performance; 3D graphics engine built on top of standard Graphics2D library.

Activities/Leadership

- 2023–Present **Captain, Former Secretary, Run Leader**, *WPI Running Club*
Organized weekly run leader schedules, planned routes and other club activities, led runs and workouts. Helped arrange club participation in NIRCA races.